Lab workbook

SMBC Data & AI Workshop

> **All workshop data is synthetic.** The labs use a made-up SMBC APAC corporate lending book: every borrower, group, limit and balance is fictional. No SMBC data, systems or customer information is used at any point.

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Before you start

1. Sign in. Open `<workspace URL>` in Microsoft Edge or Google Chrome and sign in with your SMBC account. The workshop's **Labs** folder has one notebook per session, from *00 Start here* to *04 Kit and help*; clone one (**File** → **Clone**) to run its cells.

2. Check in. Open **SQL Editor**, choose the workshop's SQL warehouse (your facilitator will name it), paste this block and click **Run**. It checks you in and creates your schema for Lab 1. Change `DA` to `DE` if you're a data engineer.

```
-- Check in and create your Lab 1 schema. Change 'DA' to 'DE' if you're a data engineer.
BEGIN
  DECLARE my_schema STRING DEFAULT regexp_replace(lower(split_part(current_user(), '@', 1)), '[^a-z0-9_]',
'_');
  INSERT INTO smbc_workshop.ops.checkin SELECT current_user(), 'DA', current_timestamp();
  CREATE SCHEMA IF NOT EXISTS IDENTIFIER('smbc_workshop.' || my_schema);
  SELECT 'smbc_workshop.' || my_schema AS your_schema;
END
```

3. Your schema. The check-in shows its name. Your Lab 1 tables go in `smbc_workshop.<your schema>`: the part of your email address before the @, in lower case, with any other characters replaced by `_`. For example, `jane.tan@…` becomes `jane_tan`.

4. Getting help. Turn your card to red, or raise a hand, and a helper will come to you.

5. Short on time? The reference tables in `smbc_workshop.gold` hold the Lab 1 result, so you can carry on with Lab 2 and the Genie Code segment whatever happens in Lab 1.

1. The scenario and the data

The labs use a synthetic **SMBC APAC** corporate lending book with about USD 27bn of drawn exposure across 12 booking branches in Asia Pacific. Every borrower, group and figure is made up. Borrowers belong to counterparty groups, such as a conglomerate and its subsidiaries in several countries, and each group has an approved credit limit set by the risk team. As at SMBC, the fiscal year runs from April to March.

The story: the Monday exposure report

It's 8:00 on Monday at the APAC hub. The Credit Committee meets at 10:00 and wants the top counterparty-group exposures across all 12 booking branches, in USD and JPY. Behind it is the question the CDAO asks the room at the start of the day: "*Which client groups are closest to their limits, and why?*"

Why that's hard today. A group's exposure is spread across subsidiaries in several countries, booked in different branches and currencies, so no single system adds it up. Answering means four system extracts, the risk team's spreadsheet of limits and days of reconciliation, and the extracts have the problems in the notes below. A group can creep towards its limit without any one report showing it. And the committee asks in the bank's own terms, such as "*the end of FY2026 H1*", which a tool that assumes a calendar year gets wrong.

How you solve it, one session at a time:

Session	What you build or use	What it solves
Lab 1: automate it	A **Lakeflow Designer** pipeline from the raw extracts to your own `gold_group_exposure` table, scheduled daily with **Lakeflow Jobs**	One trusted figure per group and month: duplicates removed; currencies, dates and FX rates fixed; bad rows held back with a reason; every step visible as code and as lineage
Lab 2: make it self-service	An **AI/BI dashboard** for the Credit Committee and a **Genie Agent**, shared through **Genie One**	The questions the committee always asks, on one page: total exposure, the largest groups and how close each is to its limit, the trend, and a watchlist. New questions answered in plain English or Japanese, using the April-to-March fiscal year you teach the agent, with the SQL behind every answer
Genie Code: find out why	**Genie Code** in a notebook	What changed since April; how much of a group's rise is currency movement rather than new lending; and a legacy SQL procedure rewritten for Databricks, with you in control of what runs
All day: governance	**Unity Catalog**, under everything you build	Lineage, data-quality checks and permissions that follow the data: a Genie Agent shows each person only the branches and borrower names they're allowed to see

By the end of the day you can answer the CDAO's question yourself, with numbers you can trace back to the source files. The quiz checks what you found.

Source files (`/Volumes/smbc_workshop/raw/landing/`)

Folder	Format	Rows	What it is
`counterparties/`	CSV	2,370	Customer master: borrower, parent group, country, sector, internal rating. Names in English, plus local script where available (日本語, 中文, ไทย, 한국어, Tiếng Việt).
`facilities/`	JSON Lines	6,047	Credit facilities: product, currency, committed amount, start and maturity dates, booking branch, collateral.
`monthly_balances/`	CSV	74,799	Month-end drawn balance and days past due per facility, April 2025 to September 2026.
`fx_rates/`	CSV	252	Month-end FX rates to USD and JPY for 14 currencies.

Folder	Format	Rows	What it is
`group_limits/`	Excel	420	The risk team's spreadsheet of approved limits per counterparty group.
`sector_emission_factors/`	CSV	16	Illustrative emission factors by sector, for an optional ESG stretch question.

Notes from the source-system owners

These are the kind of handover notes a data team gets, and Lab 1 deals with each of them:

- "*The customer master extract sometimes contains the same row twice.*"
- "*Currency codes on facilities are typed by hand, so expect `usd`, `US$`, `S$` and `RMB`.*"
- "*The trade-finance system writes dates as DD/MM/YYYY; the core loan system uses YYYY-MM-DD.*"
- "*Bangkok re-sent its September trade-finance extract. Keep the latest copy of each row (see `extract_ts`).*"
- "*FX rates are captured on the last business day of each month, not always the calendar month end.*"
- "*A few facilities were booked before the customer record was created in the master.*"

Reference tables (`bronze`, `silver` and `gold`)

The reference solution to Lab 1, in the medallion layout. Anyone can use the gold tables to carry on with Lab 2 and the Genie Code segment.

Table	Grain	Use
`gold.group_exposure`	Counterparty group × month	Exposure, limit and `utilisation_pct` per group
`gold.facility_exposure`	Facility × month	Exposure with borrower, country, sector, branch, product, currency and days past due
`silver.*`, including `silver.quarantine_facilities`	Cleaned inputs, and the rows held back with a reason	For comparing with your own Lab 1 output
`bronze.exposure_staging`	Balance × facility × customer, duplicates included	The extracts joined and converted to USD and JPY, before any data-quality checks: where Lab 1 starts
`bronze.*`	The extracts as delivered, as tables	Where the cleaning starts

2. Lab 1: Build the trusted exposure table

Goal: turn the raw extracts into `gold_group_exposure`, a table the Credit Committee can trust. You remove the duplicates, quarantine the bad rows, roll up to each counterparty group, and publish the result as a daily job, with every step visible as code and as lineage.
Time: 20 minutes: a 15-minute core path, then a 3-minute facilitator close.
Tools: Lakeflow Designer, Lakeflow Jobs, Unity Catalog.

> **Short on time?** Nothing in Lab 2 depends on finishing Lab 1: the gold reference tables have the same result. Work in pairs, ideally one analyst and one engineer, with the engineer driving.

Core path

You start from `smbc_workshop.bronze.exposure_staging`, which the workshop's daily job builds from the extracts every morning. It joins the balances to their facilities, customers and FX rates. It cleans the currency codes and both date formats, and converts to USD and JPY. Nobody has checked it, though. The problems in the source-system owners' notes are still in it, so its total is too high. Making it trustworthy is your job.

1. Start and bring in the data (3 min).

- Click **+ New** → **Visual data prep**. Rename the tab to `Monday exposure report`. In the compute picker at the top right, choose **Serverless**.
- In the canvas toolbar, open **Preview** and select **All rows**. By default, previews process only the first 1,000 rows, and the row counts in this lab are for all rows.
- Add a **Source** operator → **Browse existing** → `smbc_workshop` → `bronze` → **`exposure_staging`**. The preview shows 76,166 rows: one for each balance, with its facility, customer, group, and exposure in USD and JPY.

2. Fix the data quality (7 min). Add these two operators, the first after the Source and the second after the first:

Add	Settings	Check
Unique	**Unique by: Selected columns** `facility_id` and `as_of_date`, and **Order by** `extract_ts` descending, so that the latest copy of each row is kept	76,166 → 74,757 rows: one for each facility and month end. This removes Bangkok's re-sent extract and the extra rows from duplicate customer records
Guardrails	**Expression** checks `committed_amount > 0` and `maturity_date >= start_date` (not **Value range**, which includes 0), and a **Not null** check on `parent_group_id`. **When a check fails: Warn and continue with passed rows**. Connect **Failed** to an **Output** that writes the table `quarantine_facilities` to your schema	474 failed rows, from 36 facilities: 12 with a negative commitment, 9 that mature before they start, and 15 whose customer isn't in the master yet. The **Errors** output lists the checks that failed. That's your data-quality evidence

3. Roll up and publish (5 min).

- After Guardrails' **Passed** output, add **Aggregate**. **Group by** `as_of_date`, `parent_group_id` and `group_name`. Under **Aggregate by**, add the **SUM** of `exposure_usd`, named `exposure_usd`, and the **SUM** of `exposure_jpy`, named `exposure_jpy`. That gives 7,174 rows, one per group and month.
- After the Aggregate, add an **Output**: **Table** `gold_group_exposure` in your schema (the one your check-in created), **Write mode: Overwrite**.
- Click **Run all** in the canvas toolbar. It runs every operator on all rows and writes both of your Outputs.
- Click **Schedule** in the header and set the visual data prep to run every day at 07:00, Singapore time, on Serverless compute.
- Open **Table of contents** in the left panel, select an operator, and expand **Generated code**. Every visual step is real code you can review and version.

✅ **Checkpoint:** your row on the progress board turns green. To check your result against the reference, run both queries (notebook *01* in the Labs folder works out your schema for you); the two totals should be identical:

```
SELECT round(sum(exposure_usd) / 1e6, 1) AS mine_usd_m
FROM smbc_workshop.<your_schema>.gold_group_exposure WHERE as_of_date = '2026-09-30';

SELECT round(sum(exposure_usd) / 1e6, 1) AS reference_usd_m
FROM smbc_workshop.gold.group_exposure WHERE as_of_date = '2026-09-30';
```

Stretch (pick any)

- **Compare with limits:** add a **Join** with `smbc_workshop.silver.group_limits` on `parent_group_id = group_id`. Then add a **Prepare** step with a **Formula** action that creates `utilisation_pct`: `ROUND(100 * exposure_usd / group_limit_usd, 2)`.
- **Start from the raw files:** build the staging step yourself, from the extracts in `smbc_workshop.raw.landing`. A.2 has the formulas and join conditions.
- **Version and diff:** open the version history (right pane) to see a visual diff of your changes. Then commit the visual data prep to a Git folder.
- **Engineer's path:** rebuild the facility cleaning in the **Lakeflow Pipelines Editor** with Genie Code, using expectations (`CONSTRAINT valid_commitment EXPECT (committed_amount > 0) ON VIOLATION DROP ROW`). Compare the data-quality metrics with your Guardrails output.
- **Alerting:** add a task that checks "*any group above 90% utilisation?*" and sends a notification when it's true.
- **Document it:** in Catalog Explorer, accept or edit the AI-suggested description for `gold_group_exposure`. Better metadata means better Genie answers in Lab 2.

3. Lab 2: Credit Committee in 30 seconds

Goal: give the Credit Committee a dashboard for the questions they always ask, and a Genie Agent for the ones they haven't asked yet. Teach the agent the bank's rules, and test that it's right.
Time: 40 minutes, made up of a 25-minute core path and 15 minutes of stretch exercises.
Tools: AI/BI dashboards, Genie Agents, Genie One, Unity Catalog.

> Use the **gold** reference tables (`smbc_workshop.gold.group_exposure` and `smbc_workshop.gold.facility_exposure`) so everyone's answers match. If you finished Lab 1, your own tables give the same numbers.

Core path

1. Build the dashboard (10 min).

- Click **+ New** → **Dashboard**. On the **Data** tab, add both gold tables.
- On the canvas, ask Genie Code: "*Bar chart of the top 10 counterparty groups by exposure_usd at the latest as_of_date, coloured by utilisation_pct.*"
- Add these widgets:
  - a **counter** for total exposure at the latest month end;
  - a **counter** for groups above 80% utilisation;
  - a **line chart** of total exposure by month;
  - a **bar chart** of exposure by country.
- Add filters for `as_of_date` and `booking_branch`, then **Publish**.
- Open `<workspace URL>/one`: this is Genie One, what a business user sees.

2. Create a Genie Agent (5 min).

- Click **Genie Agents** → **New**, select the two gold tables (`group_exposure` and `facility_exposure`), and click **Create**.
- Genie Code reads the tables and suggests descriptions and example queries. Review them, and accept the ones that are right.
- In **Configure** → **Settings**, set the title to `Credit Risk — <your name>` and the default warehouse to the SQL warehouse you checked in with.
- Add the starter instructions (A.3), the two example SQL queries (A.5) and the common questions (A.6).

3. Ask, and check the SQL (5 min). Try "*Which counterparty groups are closest to their limits?*" and "*Show exposure in yen by booking branch at the latest month end.*" Open the generated SQL for each answer. Trust, but verify.

4. The fiscal-year test (4 min). Before adding the fiscal-year instruction, ask:

- "*What was Orchid Group's utilisation at the end of Q1 FY2026?*"
- "*Show Orchid Group's utilisation for each month of FY2025.*"

Open the SQL and check which dates Genie used. Then add the fiscal-year instruction (A.4) and ask both again.

✅ **Checkpoint:** ask "*Which groups were above 90% of their limit at the end of FY2026 H1?*", then submit the name of the group:

```
INSERT INTO smbc_workshop.ops.lab_answers SELECT current_user(), 'lab2', '<group name>',
current_timestamp();
```

Stretch (pick any)

- **Stump the Genie:** find a question it gets wrong, fix it with an instruction or an example query, and ask again. Share your best stump-and-fix with a facilitator: there's a prize.
  - *Starter example (tested):* ask "*Which booking branch has the highest share of trade finance in arrears?*" and open the SQL. In one test run Genie counted facilities; in another it weighted by exposure and asked which one we meant. Add the instruction "*Shares of arrears are measured by exposure (exposure_usd), not by number of facilities*", and the answer becomes consistent.
- **Benchmarks:** add the benchmark questions (A.7) with their expected SQL and run an evaluation. Benchmarks are how a team shows a Genie Agent is accurate before sharing it.
- **Agent mode:** ask "*Why has the most utilised group's limit usage risen since April 2026?*" and watch it plan, run several queries and write a report.
- **In Japanese:** 「2026年度上期末時点で、与信限度額の90%を超えている取引先グループはどこですか？」 Answers may sometimes come back in English.
- **Share it:** attach your Genie Agent to your dashboard, or schedule a Monday 08:00 PDF subscription of the dashboard.
- **Semantic layer:** use the agent's ⋮ menu → **Export to metric view** to turn the definitions you taught it into a governed metric view.

4. AI-assisted development with Genie Code

15 minutes, mostly a live demo; participants can follow along in notebook *03* in the Labs folder, which has the prompts (A.8) and the legacy procedure (A.9).

1. **Explore:** profile `gold.group_exposure` and ask what is unusual about the latest month compared with April.
2. **Generate:** split one group's exposure change since April into an **FX effect** and a **new-drawdowns effect**, by subsidiary, and chart it. This answers the "why" behind the Credit Committee's question.
3. **Explain and convert:** paste a legacy Synapse T-SQL procedure (A.9). Genie Code explains what it does, points out logic risks, and rewrites it as Databricks SQL against the workshop tables.
4. **Iterate safely:**
   - Show the action-approval setting: by default an AI classifier approves actions within your request and blocks actions beyond it, or you can approve each one yourself.
   - Ask for an explanation with "*don't run any code*".
   - Show that Genie Code can only reach data your Unity Catalog permissions allow.
   - Roll back a change with revision history.

5. Resources

Resource	Link
Free self-paced training and learning paths for data analysts and data engineers	https://www.databricks.com/learn/training
Certifications (Data Analyst Associate, Data Engineer Associate)	https://www.databricks.com/learn/certification
Databricks Free Edition, for practice outside work	https://www.databricks.com/learn/free-edition
Free virtual workshops (4 Nov *Hands-on with Genie*, 12 Nov *AI-Powered Data Engineering with Lakeflow*, 18 Nov *Build, Orchestrate and Improve Agents*; Americas time slots, recordings available)	https://www.databricks.com/learn/training/workshop
Lakeflow Designer documentation	https://learn.microsoft.com/azure/databricks/designer/what-is-lakeflow-designer
Genie Agents documentation	https://learn.microsoft.com/azure/databricks/genie-agents/
Genie One documentation	https://learn.microsoft.com/azure/databricks/genie-one/
Genie Code documentation	https://learn.microsoft.com/azure/databricks/genie-code/

Appendix A: Copy-paste kit

A.1 Check-in and checkpoint queries

```
-- Check in and create your Lab 1 schema. Change 'DA' to 'DE' if you're a data engineer.
BEGIN
  DECLARE my_schema STRING DEFAULT regexp_replace(lower(split_part(current_user(), '@', 1)), '[^a-z0-9_]',
'_');
  INSERT INTO smbc_workshop.ops.checkin SELECT current_user(), 'DA', current_timestamp();
  CREATE SCHEMA IF NOT EXISTS IDENTIFIER('smbc_workshop.' || my_schema);
  SELECT 'smbc_workshop.' || my_schema AS your_schema;
END

-- Lab 2 checkpoint
INSERT INTO smbc_workshop.ops.lab_answers SELECT current_user(), 'lab2', '<group name>',
current_timestamp();
```

A.2 Lab 1 settings (Designer)

The Unique (step 2): Selected columns `facility_id` and `as_of_date`, with **Order by** `extract_ts` descending.

The Guardrails checks (step 2). Paste each of these into an **Expression** check, then add a **Not null** check on `parent_group_id`:

```
committed_amount > 0
```

```
maturity_date >= start_date
```

The Aggregate (step 3): Group by `as_of_date`, `parent_group_id` and `group_name`, with the **SUM** of `exposure_usd` named `exposure_usd` and the **SUM** of `exposure_jpy` named `exposure_jpy`.

For the stretch exercises:

Stretch	Operator	Expression or condition
Compare with limits	Join with `smbc_workshop.silver.group_limits`	`parent_group_id = group_id`
Compare with limits	Prepare, Formula for a new column `utilisation_pct`	`ROUND(100 * exposure_usd / group_limit_usd, 2)`
Start from the raw files	Prepare after `facilities`, Formula for `currency`	`CASE upper(trim(currency)) WHEN 'US$' THEN 'USD' WHEN 'S$' THEN 'SGD' WHEN 'RMB' THEN 'CNY' ELSE upper(trim(currency)) END`
Start from the raw files	Prepare after `facilities` for `start_date` and `maturity_date`, and after `monthly_balances` for `as_of_date`	`coalesce(to_date(try_to_timestamp(start_date, 'yyyy-MM-dd')), to_date(try_to_timestamp(start_date, 'dd/MM/yyyy')))`, with each date column in place of `start_date`
Start from the raw files	Prepare after `monthly_balances`, Formula for a new column `month`; and after `fx_rates`, one for `rate_month`	`trunc(as_of_date, 'MM')` and `trunc(CAST(rate_date AS DATE), 'MM')`. FX rates are dated on the last business day of the month
Start from the raw files	Joins	Facilities to `counterparties` on `counterparty_id`; balances to facilities on `facility_id`; `fx_rates` on `currency = currency_code` and `month = rate_month`

Stretch	Operator	Expression or condition
Start from the raw files	Prepare after the FX join	`exposure_usd = drawn_amount * usd_rate` and `exposure_jpy = drawn_amount * jpy_rate`

A.3 Genie Agent starter instructions

```
This agent answers questions about SMBC APAC's corporate lending exposure. All data is synthetic.

- Exposure means drawn balance. Use exposure_usd unless the user asks for yen; then use exposure_jpy.
- Utilisation means a counterparty group's exposure divided by its approved group limit (utilisation_pct in group_exposure). Above 90% is "near its limit"; above 100% is "in breach".
- Balances are month-end snapshots. For "current" or "latest", use the most recent as_of_date. Never add balances across months; compare month ends instead.
- A counterparty group's exposure includes all of its subsidiaries in every country (parent_group_id).
- Booking branch codes: SIN Singapore, HKG Hong Kong, BKK Bangkok, SGN Ho Chi Minh City, JKT Jakarta, BOM Mumbai, SYD Sydney, SEL Seoul, SHA Shanghai, KUL Kuala Lumpur, MNL Manila, TPE Taipei.
- Arrears means days_past_due greater than 30.
- Show amounts in USD millions with one decimal place unless asked otherwise.
```

A.4 The fiscal-year instruction (add in Lab 2, step 4)

```
The fiscal year runs from 1 April to 31 March and is named after the calendar year it starts in:
FY2026 = 1 April 2026 to 31 March 2027. H1 = April to September; H2 = October to March.
Q1 = April–June, Q2 = July–September, Q3 = October–December, Q4 = January–March.
"End of FY2026 H1" means as_of_date = 2026-09-30. "FY2026 to date" means as_of_date from 2026-04-30 onwards.
```

A.5 Example SQL queries

"Top 10 counterparty groups by exposure at the latest month end"

```sql
SELECT group_name, round(exposure_usd / 1e6, 1) AS exposure_usd_m, utilisation_pct
FROM smbc_workshop.gold.group_exposure
WHERE as_of_date = (SELECT max(as_of_date) FROM smbc_workshop.gold.group_exposure)
ORDER BY exposure_usd DESC
LIMIT 10;
```

"Exposure by country and sector at a given month end" (parameterised)

```sql
SELECT country_code, sector, round(sum(exposure_usd) / 1e6, 1) AS exposure_usd_m
FROM smbc_workshop.gold.facility_exposure
WHERE as_of_date = :as_of_date
GROUP BY country_code, sector
ORDER BY exposure_usd_m DESC;
```

A.6 Common questions

```
Which counterparty groups are closest to their limits?
How has total exposure changed month by month since April 2025?
What is our exposure to Vietnam by sector at the latest month end?
Show exposure in yen by booking branch at the latest month end.
Which booking branch has the highest share of trade finance in arrears?
```

A.7 Benchmark questions (stretch)

Question	Expected SQL
Total exposure in USD at 30 Sep 2026	`SELECT sum(exposure_usd) FROM smbc_workshop.gold.facility_exposure WHERE as_of_date = '2026-09-30'`
Which groups were above 90% of their limit at the end of FY2026 H1?	`SELECT group_name, utilisation_pct FROM smbc_workshop.gold.group_exposure WHERE as_of_date = '2026-09-30' AND utilisation_pct > 90`
Top 5 groups by exposure at the end of June 2026	`SELECT group_name, exposure_usd FROM smbc_workshop.gold.group_exposure WHERE as_of_date = '2026-06-30' ORDER BY exposure_usd DESC LIMIT 5`
Utilisation by month in FY2026 for the most utilised group	`SELECT as_of_date, utilisation_pct FROM smbc_workshop.gold.group_exposure WHERE parent_group_id = (SELECT parent_group_id FROM smbc_workshop.gold.group_exposure WHERE as_of_date = '2026-09-30' ORDER BY utilisation_pct DESC LIMIT 1) AND as_of_date >= '2026-04-30' ORDER BY as_of_date`
Share of SGN trade-finance exposure in arrears at 30 Sep 2026	`SELECT 100 * sum(CASE WHEN days_past_due > 30 THEN exposure_usd ELSE 0 END) / sum(exposure_usd) FROM smbc_workshop.gold.facility_exposure WHERE as_of_date = '2026-09-30' AND booking_branch = 'SGN' AND source_system = 'TRADE_FIN'`

A.8 Genie Code prompts (AI-assisted development segment)

```
Profile smbc_workshop.gold.group_exposure and tell me what is unusual about the latest month compared with April 2026.
```

```
For the group with the highest utilisation at 30 Sep 2026, split the change in USD exposure between 30 Apr 2026 and 30 Sep 2026 into an FX effect and a new-drawdowns (volume) effect, by subsidiary, using smbc_workshop.gold.facility_exposure. Use: volume = (Sep balance − Apr balance) × Apr rate;
FX = Sep balance × (Sep rate − Apr rate). Chart it.
```

```
Explain what this procedure does and point out any risks in its logic. Then rewrite it as Databricks SQL against the smbc_workshop.gold tables. Just explain first, don't run any code.
<paste A.9>
```

A.9 Legacy Synapse procedure (synthetic example)

```
-- Legacy month-end group exposure report (synthetic example for the workshop)
CREATE PROCEDURE rpt.usp_GroupExposure_Monthly @AsOfDate DATE
AS
BEGIN
    SET NOCOUNT ON;

    IF OBJECT_ID('tempdb..#bal') IS NOT NULL DROP TABLE #bal;

    SELECT b.facility_id,
           b.drawn_amount,
           ISNULL(f.currency, 'USD') AS ccy,
           cp.parent_group_id
    INTO #bal
    FROM stg.monthly_balances b
    INNER JOIN stg.facilities f ON f.facility_id = b.facility_id
    INNER JOIN stg.counterparties cp ON cp.counterparty_id = f.counterparty_id
    WHERE b.as_of_date = @AsOfDate
      AND f.committed_amount > 0;

    SELECT TOP 20
           g.group_name,
           SUM(b.drawn_amount * fx.usd_rate) AS exposure_usd,
           CAST(100.0 * SUM(b.drawn_amount * fx.usd_rate) / NULLIF(g.group_limit_usd, 0) AS DECIMAL(9, 2))
               AS utilisation_pct,
           CASE WHEN SUM(b.drawn_amount * fx.usd_rate) > 0.90 * g.group_limit_usd THEN 'RED'
                WHEN SUM(b.drawn_amount * fx.usd_rate) > 0.75 * g.group_limit_usd THEN 'AMBER'
                ELSE 'GREEN' END AS rag_status
    FROM #bal b
    CROSS APPLY (SELECT TOP 1 r.usd_rate
                 FROM ref.fx_rates r
                 WHERE r.currency_code = b.ccy AND r.rate_date <= @AsOfDate
                 ORDER BY r.rate_date DESC) fx
    INNER JOIN ref.group_limits g ON g.group_id = b.parent_group_id
    GROUP BY g.group_name, g.group_limit_usd
    ORDER BY exposure_usd DESC;
END
```

Appendix B: Troubleshooting

Symptom	Likely cause	Fix
Genie Agents is in the sidebar but agents won't open	Partner-powered AI features are off	Ask a facilitator
Genie Code prompts, or a Genie Agent's Agent mode, are unavailable	Partner-powered AI features off, or cross-Geo processing enforced	Ask a facilitator
The check-in fails with a permission error, or you can't see the `landing` volume	No access to `smbc_workshop` yet	Ask a facilitator
Output fails with a permission error	Writing outside your own schema	Set **Output location** to `smbc_workshop.<your_schema>`
Row counts don't match the lab, for example the Source shows 1,000 rows instead of 76,166	The preview processes only the first 1,000 rows	In the canvas toolbar, set **Preview** to **All rows** (Lab 1, step 1)
Unique leaves more than 74,757 rows	It's set to **All columns**, or to other columns: rows from the re-sent extract differ in `extract_ts`	**Selected columns** `facility_id` and `as_of_date`, ordered by `extract_ts` descending (Lab 1, step 2)
Your total is higher than the checkpoint	The Aggregate doesn't read Guardrails' **Passed** output, or the Unique isn't set up as in step 2	Connect the Aggregate to **Passed**, and check the Unique (Lab 1, steps 2 and 3)
Your total is much lower than the checkpoint	The Aggregate reads Guardrails' **Failed** output	Connect it to **Passed** (Lab 1, step 3)
Genie picks the wrong dates for "Q1 FY2026", "FY2025" or "FY2026 H1"	It assumed a calendar year (it does this inconsistently)	Add the fiscal-year instruction (A.4)
Genie's arrears "share" is much higher than the dashboard's	It counted facilities instead of weighting by exposure	Add an instruction that shares are by exposure (see Stump the Genie)
A question in Japanese gets an English answer	Possible: the agent framework wraps prompts in English. In testing, the Japanese question in Lab 2 was answered correctly, in Japanese.	Add Japanese metadata or instructions if needed
A dashboard subscription email doesn't arrive	Mail gateway filtering	Ask a facilitator